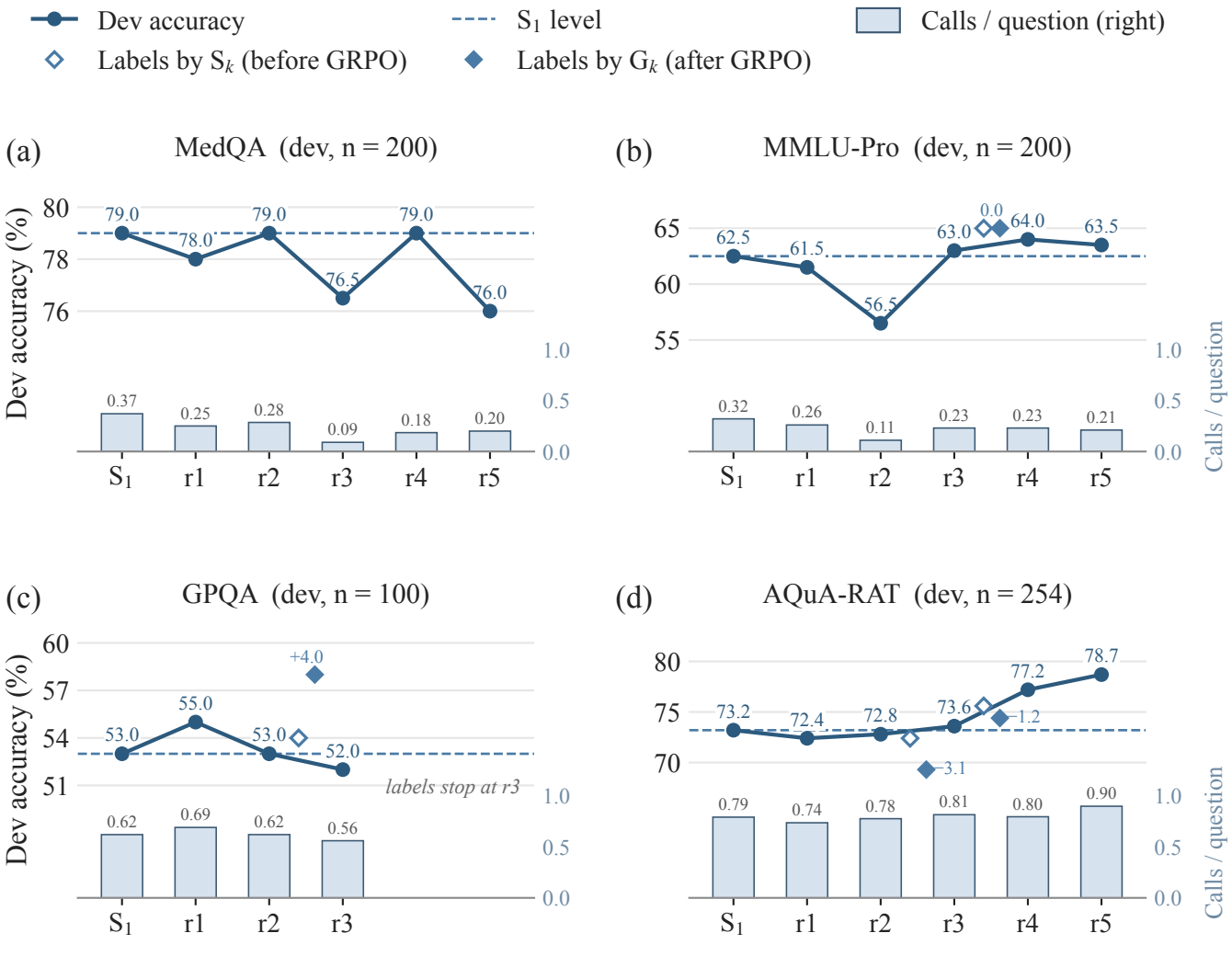
Dev accuracy
S_1 level
Calls / question (right)
Labels by S_k (before GRPO)
Labels by G_k (after GRPO)
(a)
MedQA (dev, n = 200)
Dev accuracy (%)
80
78
76
79.0
78.0
79.0
76.5
79.0
76.0
1.0
0.5
0.0
0.37
0.25
0.28
0.09
0.18
0.20
S_1
r1
r2
r3
r4
r5
(b)
MMLU-Pro (dev, n = 200)
65
60
55
62.5
61.5
56.5
63.0
0.0
64.0
63.5
1.0
0.5
0.0
Calls / question
0.32
0.26
0.11
0.23
0.23
0.21
S_1
r1
r2
r3
r4
r5
(c)
GPQA (dev, n = 100)
Dev accuracy (%)
60
57
54
51
53.0
55.0
53.0
+4.0
52.0
labels stop at r3
1.0
0.5
0.0
0.62
0.69
0.62
0.56
S_1
r1
r2
r3
(d)
AQuA-RAT (dev, n = 254)
80
75
70
73.2
72.4
72.8
73.6
77.2
78.7
−1.2
−3.1
1.0
0.5
0.0
Calls / question
0.79
0.74
0.78
0.81
0.80
0.90
S_1
r1
r2
r3
r4
r5